

KinSync Feature Map:

what's built, what's next

KinSync quietly learns an elderly person's normal phone rhythm on their own phone, and alerts family when that rhythm goes silent. Raw activity data never leaves the phone. This map shows every planned feature against the IDP review schedule.

WE ARE HERE						
Review I	Review II	Review III	Review IV	Review V	Review VI	Report
17-21 Aug 2026	21-25 Sep 2026	12-16 Oct 2026	25-29 Jan 2027	8-12 Mar 2027	29 Mar-2 Apr 2027	2 Apr 2027
Problem, objectives, survey	~20% · design + first prototype	~30% · Phase 2	~50% · Phase 3	~80% · Phase 4	100% · Open House	Final report
5 marks · done	20 marks · done	10 marks · next	15 marks	25 marks	15 marks	10 marks

- MUST-HAVEneeded for the review demo
- NICE-TO-HAVEif time allows
- ANDROIDphone app · Sri Hasini
- BACKENDserver · Nithin
- DOCS

FR-2.1 requirement ID

Built and demoed

17 featuresPhase 1, shown at Review II (25 Sep 2026)

ANDROIDElder's phone

- ✓ Consent screen

Plain-language explanation before anything is collected.

FR-7.1 · early version
- ✓ Usage-access permission

Rationale screen, then a deep link to Android Settings.

FR-2.2
- ✓ Battery-optimisation exemption

Keeps collection alive; also asks for notification permission on Android 13+.

NFR-2
- ✓ Unlock and screen on/off capture

Runs in a foreground service so the phone maker's battery saver doesn't kill it.

FR-2.1

✓ **Restarts after reboot**

Only if the elder has already given consent.

NFR-2 · FR-7.1

✓ **On-phone database**

Events stored locally in Room. Nothing is uploaded.

FR-2.4 · NFR-1

✓ **Live event list**

Debug screen showing new unlock events within seconds.

FR-2.5 · precursor

✓ **Stop monitoring**

Revokes consent on the phone and stops collection.

FR-7.3 · partial

✓ **Backend reachability check**

Calls the server's health check on launch; HTTPS only.

NFR-4

✓ **Large-text theme**

Bigger type throughout for elderly users.

NFR-6

BACKEND

Server at kinsync.ddns.net

✓ **Hardened cloud server**

Hetzner VM, Ubuntu 26.04: key-only SSH, firewall, fail2ban, automatic security updates.

NFR-4

✓ **HTTPS with automatic certificates**

Caddy reverse proxy with Let's Encrypt.

NFR-4

✓ **API service**

FastAPI running as a system service that restarts itself on failure.

NFR-2

✓ **PostgreSQL database**

Reachable only from the server itself.

NFR-4

✓ **Health endpoints**

/health and /health/db, live.

Phase 1

✓ **Automated quality gate**

Lint, type checks, complexity limits and tests with at least 90% coverage.

NFR-7

DOCS

Project documentation

✓ **Specs, design, plan and runbooks**

Requirements, architecture, roadmap with commit numbers, operating procedures, traceability matrix.

kinsync-docs

Review III: Phase 2

8 of 8 must-haves built

12–16 Oct 2026 · collect every signal on the phone and show it

Answers the Review II panel: "which features will be implemented?" The whole phase happens on the phone and nothing new leaves it. Approved by the team on 2026-10-08. **Every must-have is built and running on the demo phone (9 Oct 2026). The five nice-to-haves moved to Phase 3, after Review III (decided 9 Oct 2026).**

MUST-HAVE

built

✓ App-usage pattern

Which apps were used and when, read every 15 minutes.

ANDROIDFR-2.2

✓ Last moved

Uses the phone's low-power "significant motion" sensor and stores only the time. No step counting.

ANDROIDFR-2.7

✓ Activity: still, walking, in vehicle

Google Activity Recognition, with a permission screen in plain language.

ANDROIDFR-2.3

✓ "My day" timeline

All signals on one 24-hour view, with a day picker. App use grouped by each time the screen was on.

ANDROIDFR-2.5

✓ Daily summary

The app's main screen: first unlock, unlock count, screen time, top apps, last moved, time per activity.

ANDROIDFR-2.5

✓ Consent screen lists every signal

The elder sees exactly what is collected before it starts.

ANDROIDFR-7.1

✓ Tests for every new collector

115 logic tests on the computer, 59 database and screen tests on the device.

ANDROIDNFR-7

✓ Feature map for the panel

This page, refreshed for the review.

DOCS

Remaining phases

29 features

Review IV to the Open House · Jan–Apr 2027

Phase 3 · Review IV

25–29 Jan 2027 · ~50% · core logic + first alert

Phase 4 · Review V

8–12 Mar 2027 · ~80% · caregiver side + validation

Phase 5 · Open House + Report

29 Mar–2 Apr 2027 · 100%

Polish and accessibility pass

MUSTANDROIDNFR-6

Baseline learning

Rolling 14-day pattern, ready after 7 days.

MUST **ANDROID** FR-3

Deviation check + "Everything OK?" nudge

Asks the elder before alarming anyone.

MUST **ANDROID** FR-4.1-4.3

Elder / Caregiver role choice

MUST **ANDROID** FR-1.1

Pairing with a 6-digit code

Up to 3 caregivers per elder.

MUST **ANDROID** **BACKEND** FR-1.2-1.4

Device registration + secure tokens

MUST **BACKEND** NFR-4

Server database tables

Elders, caregivers, pairings, heartbeats, alerts. No raw activity table.

MUST **BACKEND** FR-7.2

Heartbeat from the phone

Sends only "I'm OK" plus the expected check-in window.

MUST **ANDROID** **BACKEND** FR-7.2

Dead-man's switch

The server raises an alert when a heartbeat is missed, even if the phone is off.

MUST **BACKEND** FR-4.4

Caregiver dashboard

Status for each elder: normal, nudged or escalated.

MUST **ANDROID** FR-6.1

Acknowledge and resolve alerts

MUST **ANDROID** **BACKEND** FR-6.3 · FR-4.6

Activity trend for caregivers

Aggregated only: no timestamps or app names.

NICE **ANDROID** FR-6.2

Transparency screen

Everything collected, in plain language.

MUST **ANDROID** FR-2.5

Pause / travel mode

MUST **ANDROID** FR-2.6

View and remove caregivers; revoking consent unpairs everyone

MUST **ANDROID** **BACKEND** FR-1.5 · FR-7.3

Pilot users and threshold tuning

Fewer false alarms.

MUST G3

Battery target: under 5% extra a day

MUST **ANDROID** NFR-3

Performance evaluation

Alert delay, false-alarm rate, battery.

MUST G1 · G3

Open House demo kit

Two phones, live alert, backup video.

MUST

Final report

In the prescribed format.

MUST **DOCS**

SMS fallback for caregivers

NICE **BACKEND** FR-5.3

Push alert to caregivers

Firebase Cloud Messaging.

MUST

BACKEND

FR-4.5 · FR-5

Draft of the Phase 3 API design

Data shapes for pairing, devices and heartbeat, before the backend work starts.

NICE

BACKEND

from Phase 2

Charging events

Plugged-in and unplugged times. No permission needed.

NICE

ANDROID

FR-2.8 · from Phase 2

Call activity

Number and times of calls only, never numbers or contacts. Needs a sensitive permission.

NICE

ANDROID

FR-2.8 · from Phase 2

30-day auto-delete

Raw events older than 30 days are removed from the phone.

NICE

ANDROID

NFR-1 · from Phase 2

First battery measurement

24 hours with and without collection, compared against the 5% target.

NICE

ANDROID

NFR-3 · from Phase 2

Nightly database backups

Needed before real data exists.

MUST

BACKEND

Test plan and results

End-to-end and reliability tests.

MUST

NFR-2

The privacy promise

- Raw data (unlocks, app names, motion, calls) stays on the phone.
- Only an "I'm OK" heartbeat and the expected check-in window are sent to the server.
- The server has no table that could hold raw activity.
- No GPS location, ever.

Out of scope for version 1

- Fall detection and medical diagnosis
- Step counting and continuous GPS tracking
- Wearables, iOS, audio or video monitoring
- Multiple languages; more than 3 caregivers per elder

Where the details live

- github.com/nithinvin/kinsync-docs: specs, design, roadmap
- github.com/nithinvin/kinsync-android: phone app
- github.com/nithinvin/kinsync-api: server

Phase 3 and later are plans. Priorities may change after each review.